MSG No. 162.74 $P\bar{3}1m1'$ [Type II, trigonal]

Table 1: Wyckoff site: 1a, site symmetry: -3.m1'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: 1b, site symmetry: -3.m1'

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: 2c, site symmetry: 3.21'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,2,3,4,5,6,13,14,15,16,17,18]
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[7,8,9,10,11,12,19,20,21,22,23,24]

Table 4: Wyckoff site: 2d, site symmetry: 3.21'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,2,3,4,5,6,13,14,15,16,17,18]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[7,8,9,10,11,12,19,20,21,22,23,24]

Table 5: Wyckoff site: 2e, site symmetry: 3.m1'

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,10,11,12,13,14,15,22,23,24]
2	$[0, 0, -z]$	[4,5,6,7,8,9,16,17,18,19,20,21]

Table 6: Wyckoff site: 3f, site symmetry: ..2/m1'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,5,7,11,13,17,19,23]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,6,8,12,14,18,20,24]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,4,9,10,15,16,21,22]

Table 7: Wyckoff site: 3g, site symmetry: ..2/m1'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,5,7,11,13,17,19,23]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[2,6,8,12,14,18,20,24]

3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[3,4,9,10,15,16,21,22]

Table 8: Wyckoff site: 4h, site symmetry: 3..1'

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[1,2,3,13,14,15]
2	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	[4,5,6,16,17,18]
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	[7,8,9,19,20,21]
4	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	[10,11,12,22,23,24]

Table 9: Wyckoff site: 6i, site symmetry: ..21'

No.	position	mapping
1	$[x,\ -x,\ 0]$	[1,6,13,18]
2	$[x,\ 2x,\ 0]$	[2,4,14,16]
3	$[-2x,\ -x,\ 0]$	[3,5,15,17]
4	$[-x,\ x,\ 0]$	[7,12,19,24]
5	$[-x,\ -2x,\ 0]$	[8,10,20,22]
6	$[2x,\ x,\ 0]$	[9,11,21,23]

Table 10: Wyckoff site: 6j, site symmetry: ..21'

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{2}\right]$	[1,6,13,18]
2	$\left[x,\ 2x,\ \frac{1}{2}\right]$	[2,4,14,16]
3	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	[3,5,15,17]
4	$\left[-x,\ x,\ \frac{1}{2}\right]$	[7,12,19,24]
5	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	[8,10,20,22]
6	$\left[2x,\ x,\ \frac{1}{2}\right]$	[9,11,21,23]

Table 11: Wyckoff site: 6k, site symmetry: ..m1'

No.	position	mapping
1	$[x,\ 0,\ z]$	[1,11,13,23]
2	$[0,\ x,\ z]$	[2,12,14,24]
3	$[-x,\ -x,\ z]$	[3,10,15,22]
4	$[x,\ x,\ -z]$	[4,9,16,21]
5	$[-x,\ 0,\ -z]$	[5,7,17,19]
6	$[0,\ -x,\ -z]$	[6,8,18,20]

Table 12: Wyckoff site: 12l, site symmetry: 11'

No.	position	mapping
1	$[x,\ y,\ z]$	[1,13]
2	$[-y,\ x-y,\ z]$	[2,14]
3	$[-x+y,\ -x,\ z]$	[3,15]
4	$[x,\ x-y,\ -z]$	[4,16]
5	$[-x+y,\ y,\ -z]$	[5,17]
6	$[-y,\ -x,\ -z]$	[6,18]
7	$[-x,\ -y,\ -z]$	[7,19]
8	$[y,\ -x+y,\ -z]$	[8,20]
9	$[x-y,\ x,\ -z]$	[9,21]
10	$[-x,\ -x+y,\ z]$	[10,22]
11	$[x-y,\ -y,\ z]$	[11,23]
12	$[y,\ x,\ z]$	[12,24]